DNS Threat Detection

Two-Stage Ensemble: Supervised Domain Classification
+ Unsupervised Behavioral Anomaly Detection

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INTRODUCTION

The Problem

Why DNS?

The Problem

DNS is queried by every device on a network. Essentially it's the internet's phone book.

Attackers exploit DNS for:

- Malware communication
- DNS tunneling (data exfiltration)
- Domain generation algorithms (DGA)
- Phishing via lookalike domains

Why I Chose This

Most networks are blind to DNS-layer attacks. Firewalls block ports, not malicious domains.

Full network logging for threat detection is expensive.

Pi-hole logs every DNS query, giving us a rich, real-world dataset from a live home network.

Two models catch two threat classes: known bad domains and anomalous device behavior.

INTRODUCTION

Background

Helpful information

DNS (Domain Name System)

Translates human-readable domain names (google.com) into IP addresses. Every internet-connected device uses it constantly.

Pi-hole

A network-level DNS sinkhole that logs every DNS query on a home network. Provides ground truth data on device behavior.

Supervised Learning

Train a model on labeled examples of malicious vs. benign domains to classify new domains at inference time.

Unsupervised Anomaly Detection

Learn a baseline of 'normal' device behavior from unlabeled data. Flag queries that deviate significantly from that baseline.

INTRODUCTION

What Does Success Look Like?

Supervised F1 > 0.95

High precision and recall on the DGTA benchmark — minimize both false positives and missed malicious domains.

F1 = 0.98 ✓

Meaningful Anomaly Scores

Unsupervised model produces a bimodal score distribution — clear separation between normal and anomalous behavior.

Qualitative

Complementary Coverage

The two models flag different query subsets — proving orthogonal threat coverage across domain and behavioral signals.

Ensemble

Real-World Deployment

Ultimate goal of integration with Pi-hole FTL API

Proposed System

DATA

Datasets & Feature Engineering

Two data sources — one labeled, one not

Stage 1 — DGTA Benchmark (Supervised)

Source: Public DNS threat dataset (1.65M entries)
Labels: Malicious (1) / Benign (0)
Features: Domain string-derived

Feature	Example
len	25
entropy	3.69
num_ratio	0.04
consonant_ratio	0.52
subdomain_depth	3
longest_token	11
tld_freq	0.49

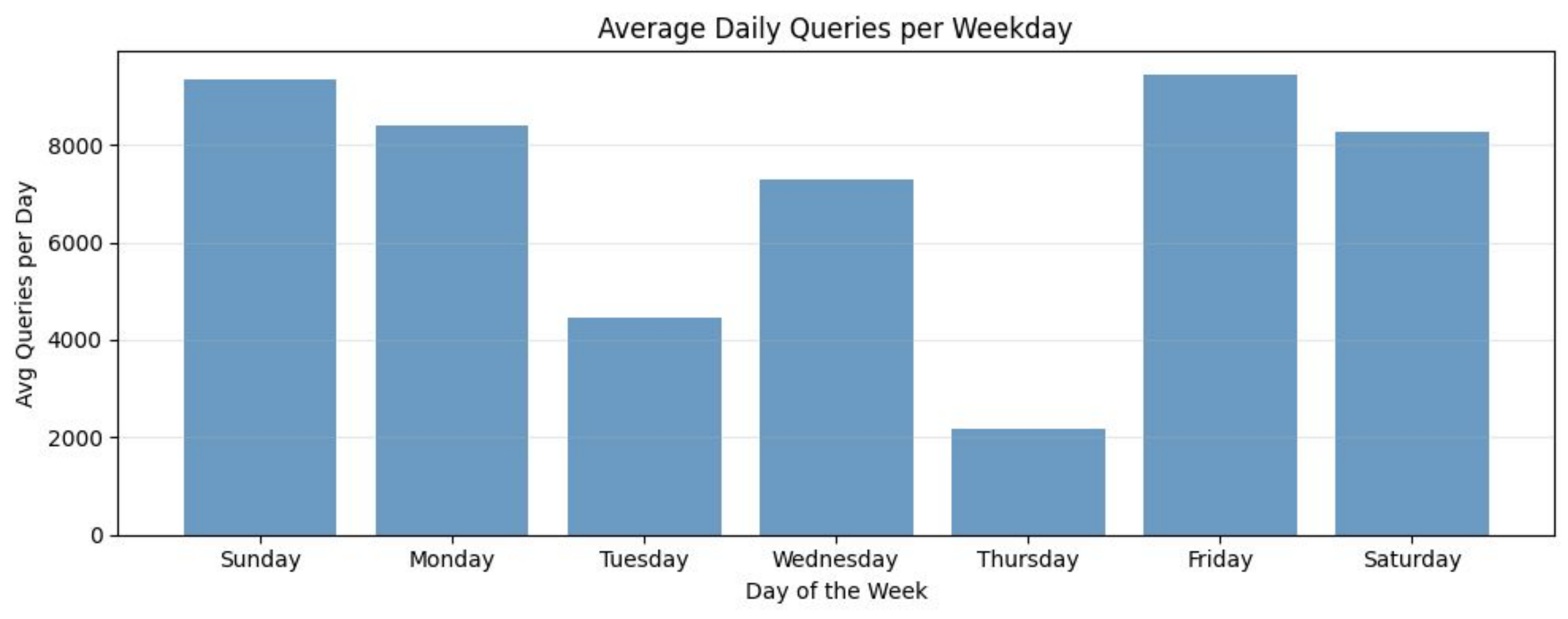
Stage 2 — Pi-hole Network Logs (Unsupervised)

Source: Live home network DNS logs (4.5M entries)
Labels: None — unsupervised
Features: Behavioral / temporal

Feature	Example
hour_sin / hour_cos	0.71 / 0.71
dow_sin / dow_cos	0.43 / −0.90
is_weekend	0
reply_time	0.0032s
status_category	allowed_cache
query_type	A
reply_type	validIP

Data Exploration

Let's get an idea of baseline behavior for one device (my desktop PC)



DATA

Data Wrangling & Cleaning

Different challenges for each dataset

Public Dataset (Supervised)

- Removed duplicate records
- Extracted TLD from domain string
- Computed tld_freq from training set distribution (saved for inference)
- Computed entropy, num_ratio, consonant_ratio, longest_token
- Stratified 70/15/15 train/val/test split

Pi-hole Network Logs (Unsupervised)

- Decoded UTF-8 errors with latin-1 fallback; dropped 2 malformed rows
- Filtered to desktop client IP; 13 complete weeks (Sun–Sat)
- Filled null reply_times: median for allowed_forwarded, p99 for in_progress
- Convert UTC to MST time
- Cyclical encoding for hour and day-of-week (sin/cos)
- Status codes mapped to 3 categories: allowed_cached, allowed_forwarded, blocked
- Query type and reply type mapped then one-hot encoded

MODELS

Model Architecture

Two complementary models — orthogonal feature spaces

Stage 1: XGBoost Classifier

Type: Gradient Boosted Trees (XGBoost)

Input: 7 domain string features

Output: P(malicious) $\in [0, 1]$

n_estimators: 300

learning_rate: 0.05

max_depth: 8

subsample: 0.8

Why XGBoost: Handles tabular domain features excellently, fast inference, built-in feature importance

Stage 2: Isolation Forest

Type: Unsupervised Anomaly Detection

Input: 21 behavioral features

Output: Anomaly score $\in (-\infty, 0.5]$

n_estimators: 300

contamination: 0.01

Preprocessing: StandardScaler via Pipeline

Why Isolation Forest: No labels required, handles high-dimensional tabular data, produces continuous anomaly scores for thresholding

TRAINING

Model Training

Feature learning and data splitting strategies

Stage 1 — Supervised

Feature Learning

All features hand-engineered from raw domain strings — no deep representation learning. XGBoost handles non-linear interactions internally.

Data Split

Set	Size	Purpose
Train	70%	Model fitting
Validation	15%	Hyperparameter tuning
Test	15%	Final evaluation

Stage 2 — Unsupervised

Feature Learning

Cyclical sin/cos encoding for time features. Status, query type, and reply type one-hot encoded. StandardScaler applied via Pipeline before Isolation Forest.

Data Split

No labeled train/test split — entire 13 weeks of data used to learn the behavioral baseline.

Evaluation is qualitative: anomaly score distribution, temporal coherence of flagged events, and cross-validation with the supervised model.

RESULTS

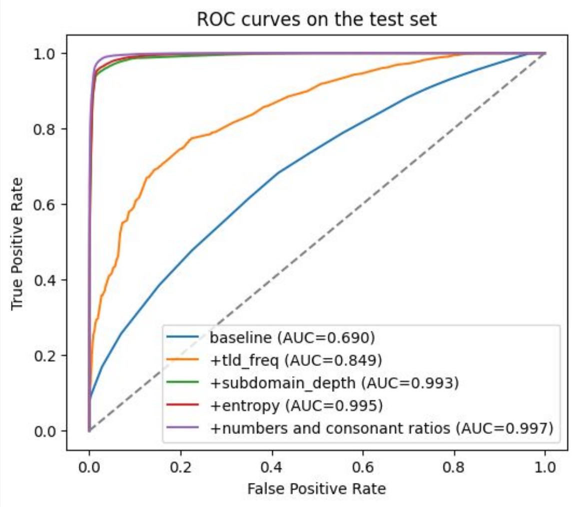
Supervised Model Performance

XGBoost on DGTA Benchmark — Test Set

97.5%	98.6%	98.0%	99.8%
Precision	Recall	F1 Score	AUC-ROC

Model Iteration History

Version	Change	F1	AUC
v1 Baseline	Only length of domain	0.370	0.69
v2	Add top level domain frequency	0.677	0.85
v3	Add subdomain depth	0.956	0.993
v4	Add entropy (how random)	0.960	0.995
v5	Add number/consonant ratios	0.971	0.997
Final	Add longest token	0.980	0.998



RESULTS

Supervised Top 20 Domains

Looks at specific flagged domains by the supervised model

	domain	sup_score	sup_prediction	STATUS
493461	95-216-39-237.top	0.998995	1	allowed_forwarded
311910	dlstreams.top	0.998176	1	allowed_forwarded
312084	joozmfyrmspbnm.com	0.997222	1	allowed_forwarded
408608	7dwbnfjp2pnmvcnrujvls5gz5oxtcribqwjtla2z0ffe77075d56ff05sac.d.aa.online-metrix.net	0.996809	1	blocked
373498	dmxtdltgndcco.site	0.996728	1	allowed_forwarded
411385	7dwbnfjp7vpmqicuknazfxowfm3nrybttb37fo2n3f589dceb7c81f20sac.d.aa.online-metrix.net	0.996425	1	blocked
409482	7dwbnfjpqwrvhpkglggfbgu6g2l3hvp6agbeklut608f72854a13fcb8sac.d.aa.online-metrix.net	0.996122	1	blocked
373503	embedsports.top	0.995671	1	allowed_forwarded
208624	steamdb.info	0.993806	1	allowed_forwarded
260515	trackcmp.net	0.993621	1	blocked
409022	7dwbnfjpd5p2sgjfkooecynqpu7i2ttdlsspfuww7ca86588c363485fsac.d.aa.online-metrix.net	0.990329	1	blocked
261312	meltonfoundation.org	0.989880	1	allowed_forwarded
11507	prodregistryv2.org	0.989811	1	allowed_forwarded
450343	browserbench.org	0.989699	1	allowed_forwarded
314268	ntvstream.cx	0.989401	1	allowed_forwarded
163933	vanillatweaks.net	0.989395	1	allowed_forwarded
398232	browser-intake-datadoghq.com	0.986330	1	blocked
69524	turbosmurfs.gg	0.986169	1	allowed_forwarded
49382	publicsuffix.org	0.984995	1	allowed_forwarded
227405	trackercdn.com	0.984147	1	allowed_forwarded

RESULTS

Unsupervised Top 20 Queries

Doesn’t look at the domain just uses context information

	domain	time	day_of_week	TYPE	REPLY	STATUS	anomaly_score
493455	how-to-pc.info	2026-04-04 09:35:14	Saturday	HTTPS	BLOB	allowed_forwarded	-0.056230
414476	gaminglaptop.deals	2026-03-25 10:54:44	Wednesday	HTTPS	BLOB	allowed_forwarded	-0.054421
415402	d6tizftlrpuof.cloudfront.net	2026-03-25 11:13:07	Wednesday	HTTPS	BLOB	allowed_forwarded	-0.052261
373434	media-delivery.cdn.com	2026-03-15 00:08:52	Sunday	HTTPS	NXDOMAIN	allowed_forwarded	-0.050557
666	_8080._https.coloradosprings.speedtest.centurylink.net	2026-01-25 00:09:09	Sunday	HTTPS	NXDOMAIN	allowed_forwarded	-0.049630
506875	stats.steaminventoryhelper.com	2026-04-04 19:15:41	Saturday	HTTPS	BLOB	allowed_forwarded	-0.049437
676	_8080._https.dende0004speedtestserver01.wiline.com.prod.hosts.ooklaserver.net	2026-01-25 00:09:09	Sunday	HTTPS	NXDOMAIN	allowed_forwarded	-0.049013
668	_8080._https.stden.highlinefast.com.prod.hosts.ooklaserver.net	2026-01-25 00:09:09	Sunday	HTTPS	NXDOMAIN	allowed_forwarded	-0.049013
448229	adimages.microcenter.com	2026-03-29 21:01:39	Sunday	HTTPS	BLOB	allowed_forwarded	-0.047650
260495	stun1.l.google.com	2026-03-01 23:55:05	Sunday	AAAA	validIP	allowed_forwarded	-0.047291
453969	tracking.marchingorder.com	2026-03-29 23:41:48	Sunday	HTTPS	BLOB	allowed_forwarded	-0.046654
534937	api.steaminventoryhelper.com	2026-04-05 23:39:36	Sunday	HTTPS	BLOB	allowed_forwarded	-0.046040
493753	mines.evaluationkit.com	2026-04-04 09:37:16	Saturday	HTTPS	CNAME	allowed_forwarded	-0.045956
311891	api.dmcfd.xyz	2026-03-07 21:51:39	Saturday	HTTPS	BLOB	allowed_forwarded	-0.044803
680	_8080._https.speedtest.den1.nitelusa.net.prod.hosts.ooklaserver.net	2026-01-25 00:09:09	Sunday	HTTPS	NXDOMAIN	allowed_forwarded	-0.044222
352934	www.envoydev.co	2026-03-14 12:52:47	Saturday	HTTPS	BLOB	allowed_forwarded	-0.044180
679	_8080._https.speedtest-den.dish-wireless.com.prod.hosts.ooklaserver.net	2026-01-25 00:09:09	Sunday	HTTPS	NXDOMAIN	allowed_forwarded	-0.043916
260497	stun.l.google.com	2026-03-01 23:55:05	Sunday	AAAA	validIP	allowed_forwarded	-0.043741
11245	_8080._https.speedtest-server-den.starry.com.prod.hosts.ooklaserver.net	2026-01-26 01:23:13	Monday	HTTPS	NXDOMAIN	allowed_forwarded	-0.043697
424692	d1cjetlwgplgi5.cloudfront.net	2026-03-28 17:51:19	Saturday	HTTPS	BLOB	allowed_forwarded	-0.043175

Unsupervised Model Results

Isolation Forest on 13 Weeks of Pi-hole Data

Evaluation Approach

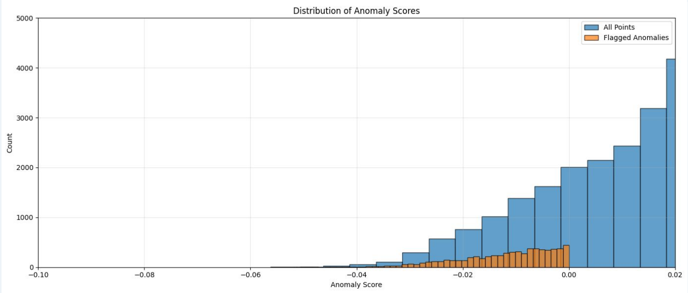
No ground truth labels → qualitative evaluation

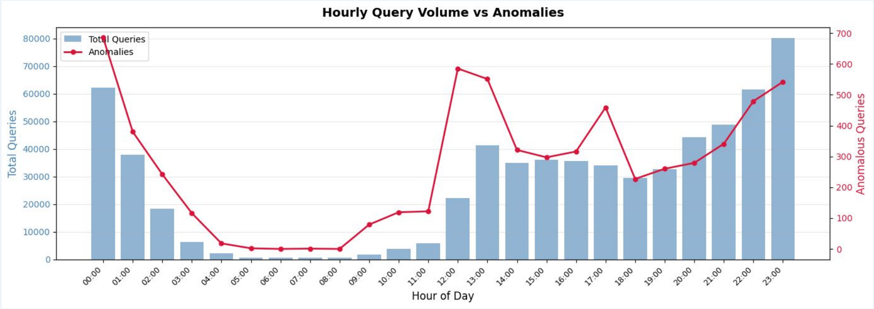
Contamination: Assumed 1% (≈ 6,500 queries flagged)

Score distribution: Strong left tail (many more benign points)

Temporal coherence: More anomalies where more queries

Cross-validation: top anomalies checked against supervised model scores for agreement.





RESULTS

Ensemble: Combined Threat Score

Merging supervised and unsupervised signals

DNS Query → XGBoost Domain Score → Isolation Forest Behavior Score → MinMax Normalize → Combined Threat Score

How the Combined Score Works

Both models score every query independently. Scores are normalized to [0, 1] via MinMaxScaler, then combined:

combined_score = 0.5 × sup_score_norm + 0.5 × unsup_score_norm

Queries flagged by both models receive the highest combined scores — representing the strongest threat signal. Queries flagged by only one model are still surfaced but at lower combined scores, allowing the analyst to prioritize.

RESULTS

Ensemble Top 20 Queries

Doesn’t look at the domain just uses context information

	domain	sup_prediction	is_anomaly	sup_score_norm	anomaly_score_norm	combined_score	STATUS
493455	how-to-pc.info	1	-1	0.948066	1.000000	0.974033	allowed_forwarded
493461	95-216-39-237.top	1	-1	1.000000	0.930244	0.965122	allowed_forwarded
450343	browserbench.org	1	-1	0.990695	0.889226	0.939961	allowed_forwarded
45185	shaderlabs.org	1	-1	0.952684	0.912772	0.932728	allowed_forwarded
502244	prismlauncher.org	1	-1	0.974572	0.885773	0.930172	allowed_forwarded
314268	ntvstream.cx	1	-1	0.990397	0.868985	0.929691	allowed_forwarded
227405	trackercdn.com	1	-1	0.985137	0.865098	0.925118	allowed_forwarded
20011	d1mybsg2h91vjj.cloudfront.net	1	-1	0.964996	0.879509	0.922252	allowed_forwarded
561407	cloudflare.com	1	-1	0.938390	0.905737	0.922064	allowed_forwarded
502248	opencollective.com	1	-1	0.936992	0.903685	0.920339	allowed_forwarded
314330	scdnmain.net	1	-1	0.965426	0.874366	0.919896	allowed_forwarded
261312	meltonfoundation.org	1	-1	0.990876	0.847900	0.919388	allowed_forwarded
75956	byronsharman.com	1	-1	0.964951	0.868896	0.916923	allowed_forwarded
13206	adoptium.net	1	-1	0.932743	0.898208	0.915476	allowed_forwarded
341043	winterparkskirental.com	1	-1	0.972202	0.848412	0.910307	allowed_forwarded
260369	bayareavolleyball.com	1	-1	0.976416	0.841280	0.908848	allowed_forwarded
260473	wisepops.net	1	-1	0.965688	0.851952	0.908820	allowed_forwarded
202098	portersfiveforce.com	1	-1	0.947652	0.859700	0.903676	allowed_forwarded
373498	dmxtdltgndcco.site	1	-1	0.997731	0.795795	0.896763	allowed_forwarded
505866	kurapixel.com	1	-1	0.890049	0.901046	0.895548	allowed_forwarded

INTEGRATION

Real-World Deployment

How this system would work in production via the Pi-hole FTL API

Pi-hole FTL REST API → **Feature Engineering** → **XGBoost + Isolation Forest** → **Alert Engine**

Pi-hole FTL API

Pi-hole v6 exposes a REST API at /api/queries that streams live DNS query data in JSON. The system polls this endpoint continuously, processing each query in real time as it arrives with no manual export needed.

Per-Device Behavioral Baseline

On first deployment, the Isolation Forest trains on each device historical query data from pihole-FTL.db. Baselines update periodically to account for evolving normal behavior without manual retraining.

Alerting and Response

Queries exceeding a combined score threshold trigger alerts via email or push notification. High-confidence hits from both models can automatically add the domain to the Pi-hole blocklist.

CONCLUSION

Challenges & Conclusion

What I learned, what I'd do differently

Evaluation Without Labels

No ground truth for the unsupervised model means success is inherently qualitative. I relied on manual interpretation and cross-model agreement.

Possible Improvement

For training the supervised portion I would consider adding a feature that compares to a dictionary of english words or common domains.

Feature Parity at Inference

TLD frequencies had to be saved from training and reloaded at inference. This means the frequencies are indicative of training data not my home traffic.

Key Takeaway

Supervised and unsupervised models are genuinely complementary. Each catches threats the other misses, making the ensemble stronger than either model alone.

Threshold Selection

Contamination is a hyperparameter with no obvious 'right' answer. I used 1% but a real deployment would need domain-expert calibration.